Bench exercise: a mortise and tenon, and a pin that pulls

Japanese Joinery by Hand · Episode 7. Mark and cut a small mortise and tenon in softwood, fit it in small steps, then pin it with an offset hole so you can feel the pin draw the joint tight. About two hours.

> **Safety.** Clamp the work. Keep both hands behind the cutting edge, and saw only on the waste side of a line. Sharp-tool technique, and how to use these tools safely, is best learned from a teacher, at a bench, in person.

1 · Prepare

1. Two planed softwood pieces of about 60 × 70 mm, each about 300 mm long: a "post" and a "sill". (The skills-test manual uses members this size for its practice piece.)
2. Mark the faces and edges you will measure from, and the center lines, as in Episode 1.

2 · The tenon

1. On the post's end: a tenon 20 mm thick, centered, and about 40 mm deep, with its shoulder line squared all round.
2. Rip down the cheeks first, then crosscut the shoulders, keeping the kerf on the waste side. Don't saw the shoulders too deep.

3 · The mortise

1. Mark the hole on the sill from the tenon itself, not from a ruler.
2. Use a chisel a little narrower than the hole. Start a few mm inside the lines, rough it out, then pare back to the lines and check the depth.

4 · Fit

- Pare in several small steps. The width should go in with light taps over a block of wood; the thickness, by hand.
- Stop when the shoulder seats all round.

5 · The pin

1. With the joint seated, bore a square or round hole (about 12 mm) through the sill, at about half its height. Mark where it meets the tenon.
2. Take the joint apart. Bore the tenon's hole about 2 mm closer to the shoulder than your mark.
3. Reassemble and drive a hardwood pin with a slightly tapered tip. Watch the shoulder close up.

What you should have

- A joint whose shoulder is pulled tight by the pin alone, with no glue.

The offset is an example (sources give about 2 mm or about 3 mm); it isn't a rule, so keep it small. Practice sizes follow the MHLW skills-test manual's scaled exercise (R7-22), not a building rule.